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Power Spectral Density Changes in the EEG during Mental Arithmetic and Eye-Opening

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Summary. The effect of mental arithmetic (standard multiplication, with eyes closed) and eye-opening on the power spectral density of the EEG (recorded bipolarly from occipital and parietal regions) have been compared. The EEG was recorded on magnetic tape from normal subjects, (medical students) in a constant environment and subsequently analysed by an analogue computer (Noratom Instrument for Statistical Analysis and Computation, ISAC).

Absolute measurements made from 15 subjects with eyes closed have shown that peaks of power at 10 Hz vary from 9 to 46 μV^2 per Hz. Qualitatively, suppression of power at 10 Hz is greater due to eye-opening than mental arithmetic.

Analysis of pooled power spectral density functions from 11 subjects showed that the effect of eye-opening differed significantly from the effect of mental arithmetic at 2.5 and 12.5 Hz. Suppression of lower frequency alpha activity was more pronounced during calculation than during eye-opening. Compared with the eyes closed condition, eye-opening showed a highly significant suppression at 10 and 12.5 Hz (alpha activity) and from 15 to 25 Hz (beta activity). Mental arithmetic induced significant suppression at 7.5 and 10 Hz and from 15 to 30 Hz. Eye-opening enhanced lower frequency activity which calculation suppressed.

Variation was greater between first, second and third calculations in pooled power spectral densities than between corresponding eye-opening epochs. It is suggested that these differences in effects of eye-opening and mental arithmetic reflect differences in underlying cerebral activity, rather than that an underlying factor common to both effects may be orbital in origin.

Introduction

The EEG changes in man associated with increased mental activity while the eyes are closed, for example, during calculation. (Adrian and Matthews, 1934; Lorens and Darrow, 1962; Chapman *et al.*, 1962; Glass, 1964, 1967) resemble superficially the changes induced by eye-opening alone. The problem is the extent to which the EEG changes associated with eye-opening can be ascribed to increased mental or cerebral activity (in the striate cortex) or conversely the extent to which these changes and those due to calculation may reflect activity of purely orbital factors. Quantitative comparison may help to evaluate these fundamental relationships. Alpha activity is more intensely suppressed by eye-opening than

by calculation with eyes closed, which has the effect of enhancing beta activity (Mundy-Castle, 1957; Volavka *et al.*, 1967). Percent time alpha activity and the amplitude of the alpha rhythm are also reduced by calculation and affected by the order of the task and the correctness of its solution (Kreitman and Shaw, 1965; Glass, 1964, 1967).

However, during calculation, the pupils dilate (Hess and Polt, 1964); Mulholland and Evans (1966), Dewan (1967) and Lippold and Novotny (1968) have proposed that ocular movements associated with eye-opening may be important influences on alpha activity. If so, the degree of resemblance of the EEG effects of eye-opening and mental arithmetic could be a factor in determining whether eye-movement might be a contributory cause of these changes in both conditions. To investigate this problem, in alpha and other frequency bands, power spectral density measurements of the EEG were made and compared during mental arithmetic with eyes closed and during eye-opening alone. Changes in power spectral density functions have been found during continuous photic and auditory stimulation (Sato and Mimura, 1956; Mimura *et al.*, 1962; Suhara and Suzuki, 1964) and detailed spectral analysis of electrical activity from different cerebral regions (Walter *et al.*, 1966) have been reported. A preliminary account of these experiments has been given (Glass and Kwiatkowski, 1968).

Method

Environment

Environmental control was achieved by recording the EEG while each subject sat in an acoustically and electromagnetically shielded chamber mounted on rubber-in-shear, in which total (audible and non-audible) sound-pressure levels were attenuated by 54 db, and all components were joined by metal-to-metal contacts of braided monel strip. The shielded EEG cable passed through the chamber wall. Two way audio-communication was maintained between subjects and experimenters. Subjects were illuminated, from behind, by a constant source.

Experimental Procedure

Subjects were medical students at the University of Birmingham, in normal health. The channels for analysis were selected from both anterior and posterior parietal and occipital regions [bipolar electrodes C3—P3, P3—O1, C4—P4 and P4—O2 (Jasper, 1958)]. Lint covered and silver disc electrodes were applied and supraorbital electrodes used to monitor eye-movements (Lorens and Darrow, 1962).

A six-to-ten minute EEG recording was taken. Each test was carried out in 11 subjects in two sets. The first set contained three resting periods during which the eyes were closed, lasting thirty seconds, each followed by a period during which eyes were open of the same duration. The second set consisted of three similar thirty-second eyes-closed periods each followed by the presentation of three two-digit multiplications exclusively composed of digits of 5 and over (Glass, 1966).

The effect of order was balanced out by the experimental design. Power spectral densities were measured on 15 subjects in total, but the complete experiment was carried out on the 11 subjects referred to above.

EEG Recording and Analysis

The EEG was monitored by oscilloscope and recorded on magnetic tape on a seven track Ampex Portable Instrumentation recorder (SP 300) and simultaneously, graphically on a Beckman 8 Channel Dynograph or an S.E.L., UV galvanometer. One tape-recorder channel was reserved for coded or spoken information.

The signal was later played back on to a tape-loop of an analogue computer, ISAC, (Noratom, Instrument for Statistical Analysis and Computation). The output of the analyser, which was a power spectral density function was written out by a graph plotter. If an EEG channel contained evidence of excessive eye-movement artefact or electromyographic activity it was not analysed. Also, if a portion of the channel contained eye-movement artefact or electromyographic activity it was not analysed. Apart from applying these criteria, no selection of samples for analysis was made which might might have biased the results. Ten seconds of real time of EEG trace were taken at random from each condition for each subject. This was played back at twice the recording speed to improve the relative frequency resolution.

Measurement and Comparison of Power Spectral Densities

Measurements of the magnitude of the power spectral density were made at increments of 2.5 Hz from zero to 30 Hz and the effects of eye-opening and mental arithmetic with the eyes closed on the power spectral density compared at each frequency. At each frequency level, the ratio of the magnitude of the power spectral density during eye-opening to the magnitude of the power spectral density of the same frequency level during the preceding eye-closure period was found for each of the three eye-opening periods for each subject. Similarly, the ratios of the magnitudes of the power spectral densities at each frequency level during calculation with eyes closed to the preceding periods of eye-closure without calculation were found for each calculation by each subject. The difference between these ratios for eye-opening and mental arithmetic conditions and the differences of these to the resting eyes-closed condition were compared by "*t*" tests at each frequency level (Schallek *et al.*, 1968). Absolute values of spectral density at different frequencies were calculated in units of μV^2 per Hz (Walter, 1968).

Results

Observations on Individual Subjects

Comparisons by way of example of the direct effects of eye-opening and calculation on the EEG (Fig. 1 A) show that suppression of the alpha rhythm by mental arithmetic, is less intense than that induced by eye-opening. Considerable suppression at 10 Hz can be seen in the power spectral density function of one subject (Fig. 1 B), and at other frequencies. The other subject shows a characteristic bimodal peak of energy at two frequencies both within the alpha range (at 8—9 and 12—13 Hz).

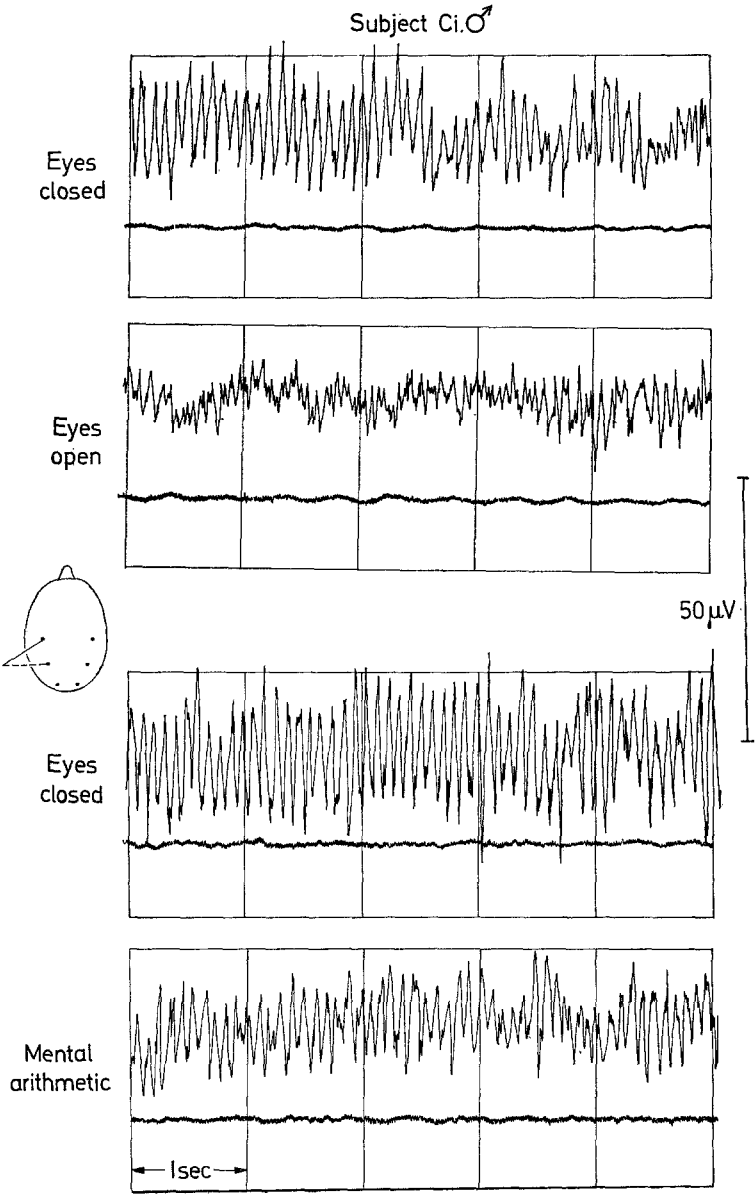

Fig. 1 A

Fig. 1. A. Traces (Ci. male 19) from (C3—P3) recorded by UV galvanometer. Lower trace in each pair is the information channel. Upper pair: eyes closed. Second pair: eyes open. (Note suppression of alpha rhythm). Third pair: eyes closed. Fourth pair: recorded (with eyes closed) during solution of two-by-one digit multiplication

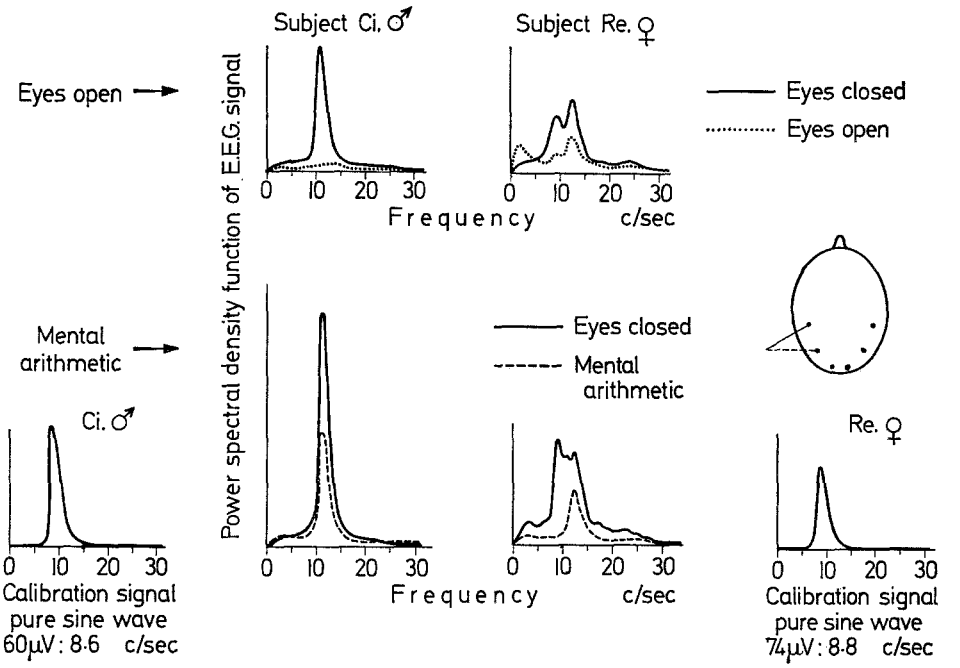

Fig. 1B

Fig. 1. B. Power spectral density functions computed by ISAC from ten second samples of EEG. On left subject Ci. (see Fig. 1A for raw traces). On the right are similar functions from another subject Re. (female). Power spectral density is shown, eyes closed (continuous line); eyes open (dotted line); and during calculation (dashed line). Note that considerable suppression of power occurs in the alpha region (subject Ci.) at 10 HZ. Subject (Re.) shows bimodal peaks, one at 8 or 9 Hz and the other at 12 or 13 Hz. Suppression of power is greater at 8 Hz than 12 Hz. Suppression of power has occurred during calculation, but not as intensely as with eye-opening. Subject Re. shows almost complete suppression at 8 Hz, greater than during eye-opening and partial suppression at 12 Hz. Note respective calibration signals on lower left and right

The effect of mental arithmetic is that (Fig. 1B) there is a high peak of energy at 10 Hz which confirms that energy suppression has been influenced by mental arithmetic but not as intensely as by eye-opening. Mental arithmetic, in the other subject, caused almost complete suppression of the energy peak at 8 Hz and a partial suppression of energy at about 12 Hz. Considerable differences in power spectral densities were found between individual subjects but individual patterns of distribution of energy were relatively constant over a period from six to ten minutes in duration (Fig. 1B).

The range of normal values of power spectral densities from five different individuals, resting with eyes closed, in ten second samples

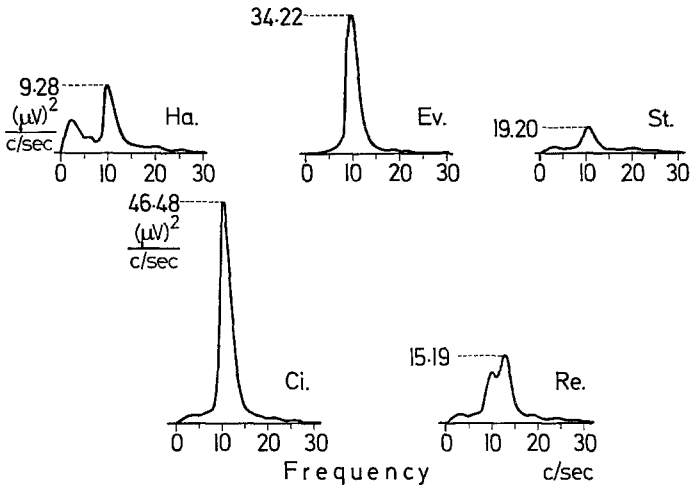

Fig. 2 shows range of normal values of power spectral densities (from same electrode positions) in five different individuals, resting with eyes closed, in 10 second samples. These cannot be drawn to the same scale. The peaks vary from 9 to 46 μV^2/Hz. Note variation in form with each subject. The absolute values are calculated in μV^2 per Hz

is shown (Fig. 2) in μV^2 per Hz (Walter, 1968). The peaks of power at 10 Hz varied from 9 to 46 μV^2 per Hz in different normal subjects.

Calculation of the ratio of differences between power spectral densities during eye-opening or mental arithmetic with eyes closed and the power spectral density with eyes closed, as a percentage of the eyes closed density at each frequency level (Fig. 3) demonstrated a suppression of power at all frequencies (0—30 Hz by eye-opening and mental arithmetic. The suppression due to eye-opening was definitely greater at 10 Hz and at higher frequencies. The second example confirms enhancement of power at low frequencies by nearly 100% by eye-opening and suppression by mental arithmetic. Suppression of power occurs at 20 Hz with eye-opening.

Quantitative Changes

The mean enhancements and suppressions of power have been calculated for eleven subjects in which data from the three eye-opening conditions and three calculation conditions have been pooled (Fig. 4). Enhancement of power in delta and low theta range (under 5.0 Hz) by eye-opening and suppresssion by mental arithmetic was found. Power at alpha frequencies around 10 Hz was suppressed during calculation but more intensely during eye-opening when characteristic suppression of

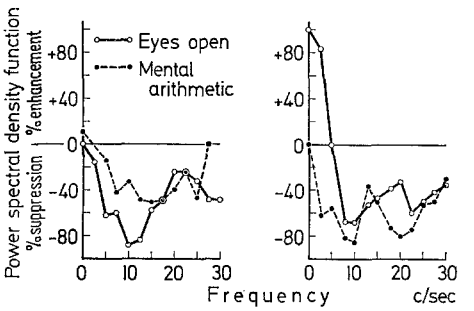

Fig. 3 is derived from measurement of height of the power spectral densities at 2.5 Hz increments from 0—30 Hz. It plots the difference between the power spectral density during eye-opening or mental arithmetic (with eyes closed) and the power spectral density with eyes closed, as a percentage of the power spectral density with eyes closed.

$$\frac{P_{\mathrm{EO/MA}} - P_{\mathrm{EC}}}{P_{\mathrm{EC}}} \times 100.$$

Data from two subjects are presented. A negative percentage indicates suppression (below line) and a positive percentage indicates enhancement of power, More intense suppression by eye-opening than calculation occurs in alpha range (left-hand graph). Considerable enhancement by eye-opening at low frequencies (delta/theta) is found (right-hand graph)

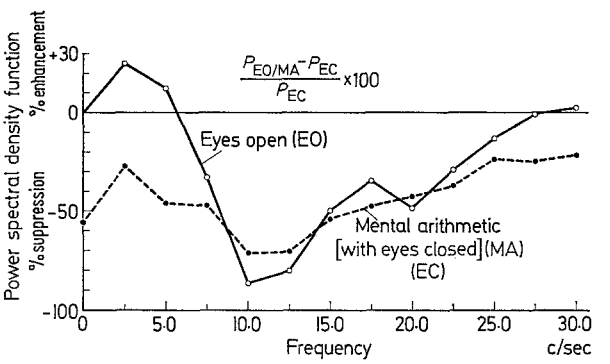

Fig. 4 shows mean enhancement and suppression (as in Fig. 3) of power pooled for 11 subjects (at 2.5 Hz intervals for eye-opening and calculation). For each subject data from three eye-opening records and three calculations have been pooled. Enhancement of power at low frequencies is seen with eye-opening in contrast to suppression by mental arithmetic. Suppression of power at alpha frequencies occurs with mental arithmetic but is more intense during eye-opening

power occurred at 20 Hz. At frequencies higher than 20 Hz the predominant effect was of suppression by both mental arithmetic and eye-opening.

Table. *Differences of ratios, EC/EO, EC/MA, EC/EC of power spectral densities at different frequencies**

Frequency levels (Hz)	EO-MA	EC-EO	EC-MA
0	−0.140	−0.38	−0.179
2.5	−0.570[b]	0.203[a]	−0.373
5.0	−0.803	0.134	−0.668
7.5	−0.372	−0.498[a]	−0.871[d]
10.0	4.790[a]	−5.827[c]	−1.032[c]
12.5	2.480[b]	−3.676[d]	−1.190[a]
15.0	−0.299	−0.987	−1.265[b]
17.5	−0.230	−0.537[b]	−0.786[c]
20.0	0.194	−0.954[c]	−0.760[c]
22.5	−0.203	−0.409[d]	−0.612[b]
25.0	−0.141	−0.309[b]	−0.450[a]
27.5	−0.327	−0.025	−0.353[b]
30.0	0.038	−0.418[a]	−0.379[d]

Probabilities; [a] 0.1—0.05; [b] 0.05—0.02; [c] 0.02—0.01; [d] 0.01—0.001; (paired "t" test, 11 subjects).

* Not more than 2 pairs of differences need be taken but 3 have been calculated to avoid recalculation.

The mean power during eye-opening and mental arithmetic, at different frequency levels, as ratios of the power with eyes closed, was compared. Statistical analysis (see Table) of the effects of eye-opening and eye-closing showed that in the lower frequencies (delta and low theta), power was enhanced by eye-opening but only at 2.5 Hz was this even doubtfully significant (mean difference of ratios: 0.203; $p = 0.1$—0.05). At 7.5 Hz, in the region of low alpha, suppression of power was doubtfully significant during eye-opening, but in the remainder of the alpha range, 10.0 and 12.5 Hz mean differences of ratios were large (5.827 and 3.676) and highly significant ($p = 0.02$—0.01 and 0.01—0.001 respectively). In the beta range, from 15 to 25 Hz, a statistically significant suppression of power occurred during eye-opening compared with the eyes closed condition.

Quantitative effects of mental arithmetic on the pooled power spectral density function were not identical with the effects of eye-opening. Up to 5.0 Hz in thetheta/delta band there was suppression of power which did not reach the level of statistical significance. Suppression of power at 7.5 Hz and 10 Hz (low alpha) was highly significant (mean difference of ratios; 0.871 and 1.032; $p = 0.01$—0.001 and 0.02—0.01) but only doubtfully significant at 12.5 Hz (high alpha). From 15 to 30 Hz (beta) (except 25 Hz) suppression of power during mental arithmetic compared with the eyes closed condition was statistically significant. Statistically significant suppression of power in the upper beta fre-

quencies at 27.5 and 30 Hz occured during mental arithmetic, in contrast to the effect of eye-opening.

Direct comparison of the effects of eye-opening with the effects of mental arithmetic showed that in the lower frequencies, only at 2.5 Hz was the suppression of power significantly greater during mental arithmetic ($p = 0.05$—0.02). At 10 Hz suppression of power by eye-opening was considerably greater than that due to mental arithmetic (mean difference of power ratios: 4.79) but only doubtfully significant ($p = 0.1$—0.05) because of the increased variance of ratios at this frequency level. At 12.5 Hz suppression by eye-opening was also greater than that due to mental arithmetic (mean difference of ratios: 2.480; $p = 0.05$—0.02).

The results of statistical analysis, therefore, have reinforced the quantitative findings and have shown a significant difference between effects of mental arithmetic and eye-opening on low frequencies of the EEG. Power at 2.5 Hz was enhanced by eye-opening and suppressed by calculation. In the alpha range, significant suppression of power occurred during both conditions, but more intensely with eye-opening. Suppression of lower frequency alpha (circa 7.5 Hz) was more pronounced during calculation than during eye-opening, and suppression of high frequency alpha (12.5 Hz) was greater during eye-opening than during calculation.

The Time Course of Power Spectral Changes

The EEG was recorded and power spectral densities measured during three periods of eye-opening and three calculations. The first, second and third periods of each condition pooled for all subjects have been examined at each frequency (Fig. 5). Suppression at 10 Hz (alpha) was less in the third eye-opening period (70%) than in the first two eye-opening periods (90%), qualitatively suggesting habituation. Suppression of power at 20 Hz occurred progressively less markedly, also suggesting habituation. Enhancement of power did not occur at low frequencies in the second of the three eye-opening periods. Each change induced by eye-opening was characteristically similar to the others, strikingly, the ratios of power at 10 and 12.5 Hz were very nearly constant at each period.

The time course of the three calculations differed qualitatively from those of eye-opening (Fig. 6). There was more variability between sucessive calculations. Suppression of power at alpha frequencies was only 55% in the first, 80% in the second and 40% in the third, where maximum suppression was at 12.5 Hz. Some increased suppression was also found at 20 Hz during the first two calculations, but, in general, suppression of power at low frequencies occurred during all calculations.

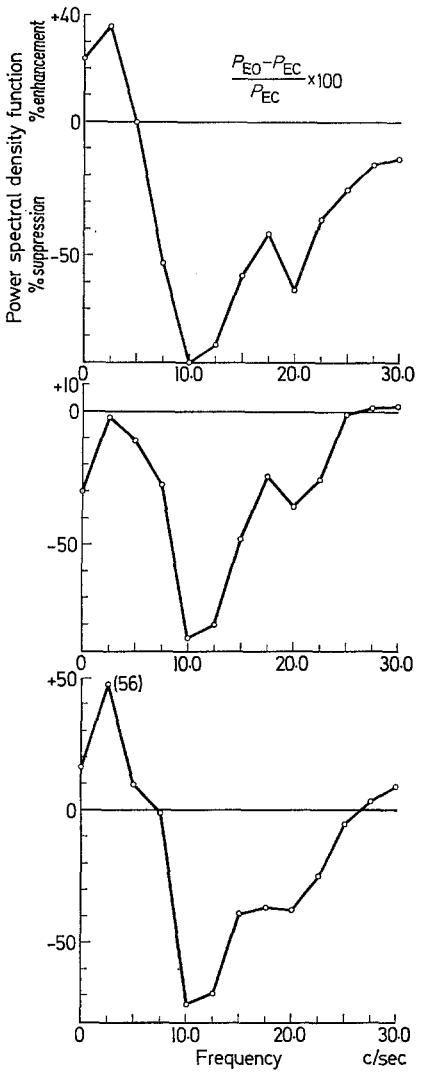

Fig. 5

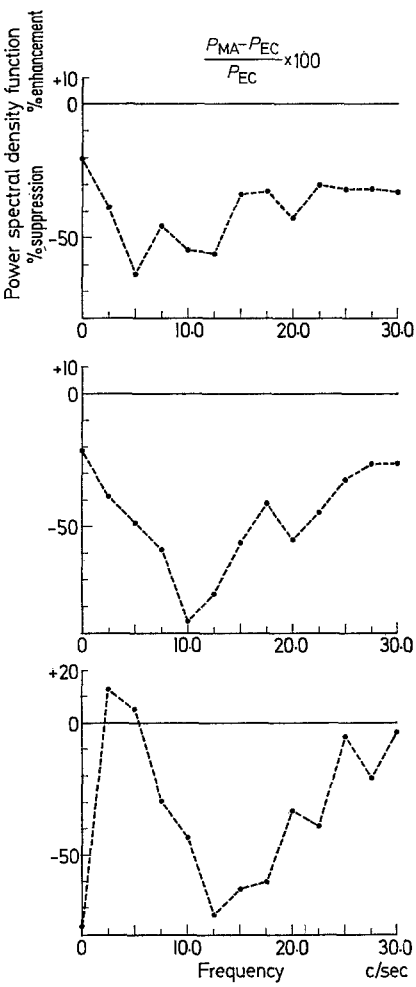

Fig. 6

Fig. 5 shows the mean power spectral density ratio changes pooled for 11 subjects with first, second and third eye-opening periods compared. (The graph baseline is at a different position in the third eye opening period)

Fig. 6. This shows power spectral density changes due to first, second and third calculations pooled for 11 subjects. These changes differ from those of eye opening (see Fig. 5)

Discussion

These results of comparison of power spectral density changes during eye-opening and calculation have demonstrated a characteristic distribution in the normal resting record for different individuals and variation of the power distribution at different times over the period studied. Eye-opening has produced a more consistent effect on power spectral density of the EEG than mental arithmetic and significantly greater suppression at 10 Hz and enhancement of power at lower frequencies than calculation. Greater suppression at 20 Hz occurred during eye-opening although, in this, the effect of sequence of presentation may be important. Mental arithmetic with eyes closed showed greater variation between the first, second and third calculations than eye-opening. The quantitative findings will be discussed, firstly, in relation to earlier work, both in power spectral density and low frequency EEG analysis, secondly, the relevance of the interpretation of the results to possible mechanisms of alpha activity, that is the relation of those EEG changes to underlying cerebral processes associated with calculation and eye-opening, will be considered and finally, the scope and limitations of power spectral analysis in this field of EEG work will be outlined.

Quantitative Findings

It might be considered advisable to assess and evaluate these quantitative findings against the background of earlier experiments in this field.

Suppression of alpha activity at 10 and 12.5 Hz was found during eye-opening and mental arithmetic, but suppression due to eye-opening was more intense. This is in agreement with the findings of Volavka *et al.* (1967) and Suhara and Suzuki (1964). Significant suppression of beta activity up to 30 Hz was found during eye-opening at different light intensities (Sato and Mimura, 1956). More augmented beta activity occurred during mental calculation than eye-opening (Mundy-Castle, 1957). In delta and low theta frequency bands, slight enhancement occurred around 2.5 Hz during eye-opening which differed significantly from suppression at this frequency associated with mental calculation.

Suppression of beta activity found during calculation and eye-opening is unlikely to be due to electromyographic activity as this only influences frequency analysis of the EEG above 27.5 Hz (Ellis and Last, 1953). Variations in the effect of beta activity during mental arithmetic may be due to differences in the problems presented or to discrete, in contrast to continuous, presentations of calculations (Sato, 1955; Mimura *et al.*, 1962).

The quantitatively significant "alpha shift" effect, that is, more pronounced suppression of the low frequency alpha during calculation than during eye-opening and a suppression of high frequency alpha greater during eye-opening than calculation, is compatible with the concept of polyrhythmic alpha activity, (Walter, 1950; Cobb, 1963). Esser and Bickford (1950) and Mimura *et al.* (1962) found that the dominant alpha frequency was depressed but that slower and also faster rhythms were augmented by continuous auditory stimuli. After repetition, these ultimately shifted back to the dominant alpha frequency of the resting condition, suggesting an incomplete habituation. Following continuous photic stimulation, Sato and Mimura (1956) found that low frequency alpha wave peaks diminish and those of high frequency are enhanced. A bimodal power spectral function of occipital alpha similar to subject Re. (see Fig. 1 B) has been described (Sato, 1955). Intermittent click stimuli enhanced the dominant alpha frequencies and some higher frequencies (Sato *et al.*, 1959) but a single flash stimulus caused temporary blocking of alpha frequencies followed by enhancement of alpha activity especially at 8.0 Hz. Bell stimulation diminished some elements and augmented others in different regions of the brain, (Mimura *et al.*, 1962).

Concerning the sequence of presentation, Kreitman and Shaw (1965) showed mean significant changes for a visual task and for continuous mental arithmetic but the effects of eye-opening and closing were not statistically significant. A significant increase in percent time alpha activity occurred during the presentation of twenty calculations (Glass, 1967). In the light of these results, our findings of no statistically significant change in the time-course of power spectral densities are not wholly expected. This difference may be due to a smaller number of calculations in the present experiment, or to the possibility that suppression of power at alpha frequencies and decline in percent time alpha may well not be exactly equivalent measures.

Relation of EEG to Underlying Cerebral Processes Associated with Calculation and Eye Opening

The contribution which these results make to the solution of the problem of the extent to which EEG changes associated with eye-opening and calculation can be ascribed to increased mental activity (in the widest sense) or cortical function, can be evaluated. In so far as eye-opening and mental arithmetic induced EEG changes that were specific to each condition, that is, which differed significantly at 2.5 and 12.5 Hz (eye-opening enhanced lower frequency activity which calculation suppressed) then a more intimate relation of these EEG changes to underlying cerebral processes is indicated. Non-specific

differences are more likely to be related to more general cerebral processes common to both mental arithmetic and eye-opening, for example, the general arousal response or orientating response (Sokolov, 1963) or perhaps associated with eye-movements or oculomotor function.

Returning to the possibility that a common origin for these changes might be found in eye-movement or orbital factors, Lorens and Darrow (1962) have shown a marked increase in eye-movements during mental arithmetic. On the hypothesis that the alpha rhythm originates as an interaction between the corneo-retinal potential and eye-muscle tremor, when the cortico-retinal fixation system is inactive (Lippold and Novotny, 1968) it might be predicted that increased eye-movement would reduce the amount of alpha activity, but it is doubtful that, with eyes shut, retinal fixation could occur during calculation. Therefore, on this hypothesis, suppression at alpha frequencies would not be anticipated during calculation, Increased eye-movement leading to increased "oculomotor" imbalance (Mulholland and Evans, 1966), would be expected to augment the amount of alpha activity rather than decrease it. Thus an enhancement rather than overall suppression of alpha activity would be predicted during calculation if the effect of calculation on the EEG were solely mediated through ocular factors. The appearance of alpha activity by the upward movement of the eyes has been explained by Dewan (1967) as a consequence of a tendency of the eyes to defocus and relax convergence. If the hypothesis that the alpha rhythm is dependent on peripheral ocular activity is valid, then the effect of increased eye-movement during calculation would be to increase not to lessen alpha activity. Mental multiplication (when eyes are open) causes dilation of the pupil (Hess and Polt, 1964) (implying defocussing and de-accommodation) which would lead to the prediction from Dewan's theory of more, not less, alpha activity during calculation.

From these results and considerations, it must be concluded that the suppression of alpha activity is an effect not of changes solely in orbital factors but a direct result of calculation or the mental activity associated with underlying cerebral or cortical processes and, in the case of eye-opening, with the functional activity associated with visual processes in the striate cortex.

Role of Power Spectral Density Function in EEG Analysis

It may be worthwhile commenting on the scope and limitations of the power spectral density function in EEG analysis in this field.

Power spectral density analysis was selected from other forms of time-series analysis because it measured energy distribution, a parameter which has already proved empirically useful in the field of the neurophysiological correlates of behaviour by graphical measurement of the EEG (Glass, 1959, 1964).

In applying a time-series analysis to the EEG, it can be assumed that the EEG signal consists of a random and a non-random component. The process giving rise to random functions is termed stationary when these functions are time-invariant. If it transpired that the random part of the EEG signal were significant then the corresponding random process should be stationary under constant conditions but should change character systematically under changing experimental conditions.

The power spectral density functions give a useful, preliminary means of description. The chief difficulty in interpretation is due to the fact that the length of an EEG record is finite whereas the mathematical description is based on infinity. With a finite length of EEG record, accuracy in a statistical sense can only be achieved if frequency resolution is sacrificed. Instead of examining power of discrete frequency components, which would require infinite time, it is necessary to measure power contained in a band of frequency, thereby performing a smoothing operation.

In this experiment, the smoothing of the power curve takes place over a range of 2.5 Hz and no finer frequency resolution should be attempted.

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